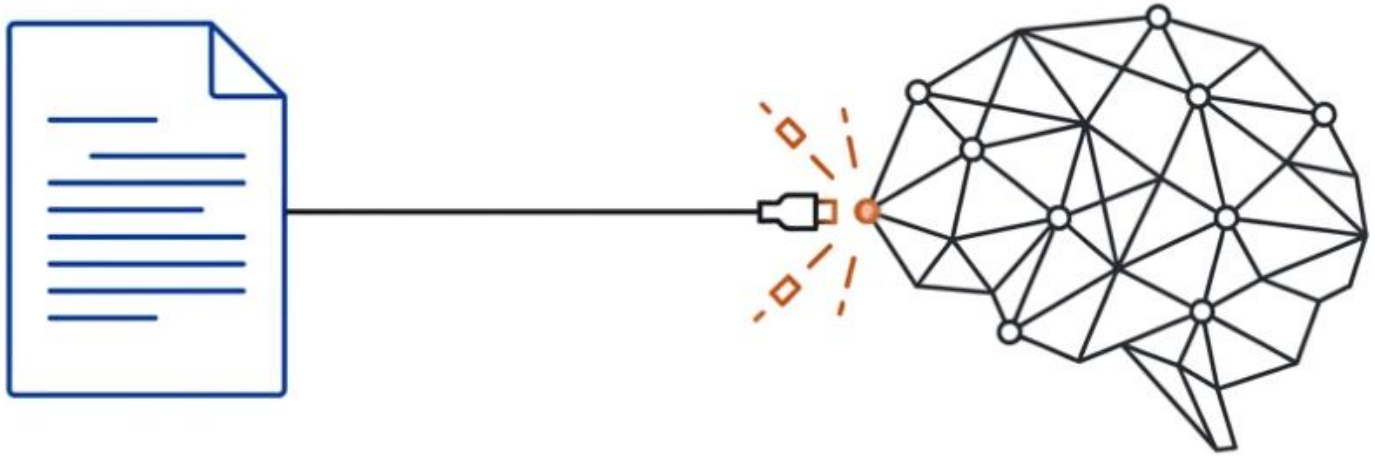
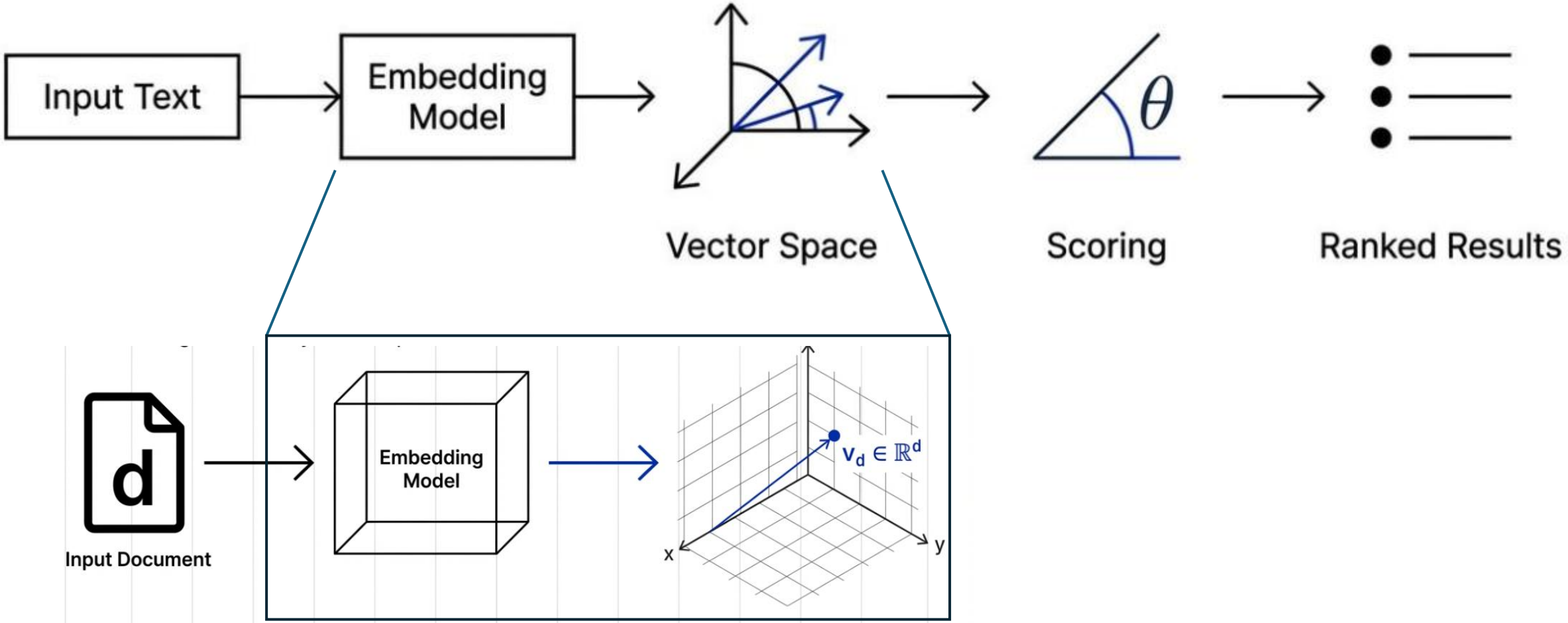


Semantic Encoder Models

Early and Late Interaction, Bi-encoder, Cross-encoder, ColBERT, and Retrieve→Rerank

https://tinyurl.com/rag-augmented

Previously...



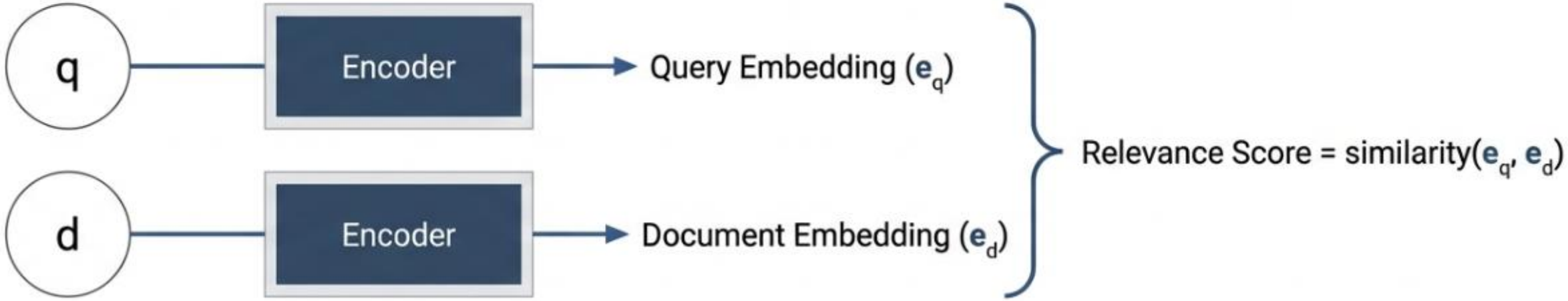
Retrieval shifts from matching keywords to finding the most similar vector in space.

Independent Encoding for Fast Retrieval

A retrieval model that encodes the query and the document independently into dense vectors.

The Scaling Advantage:
Document embeddings can be precomputed offline.
At runtime, only the query needs to be encoded.

Result: Fast, scalable retrieval for very large corpora.



Bi Encoder: Recipe

Stage 1: Language Pretraining

Method: Masked language modeling / self-supervised objectives on large text corpora.

Learns: Syntax, semantics, sentence meaning, world knowledge.

Stage 2: Retrieval Fine-Tuning

Method: Supervision using relevance datasets.

Data Structure: (q, d+) or (q, d+, d-)

The Data Fuel

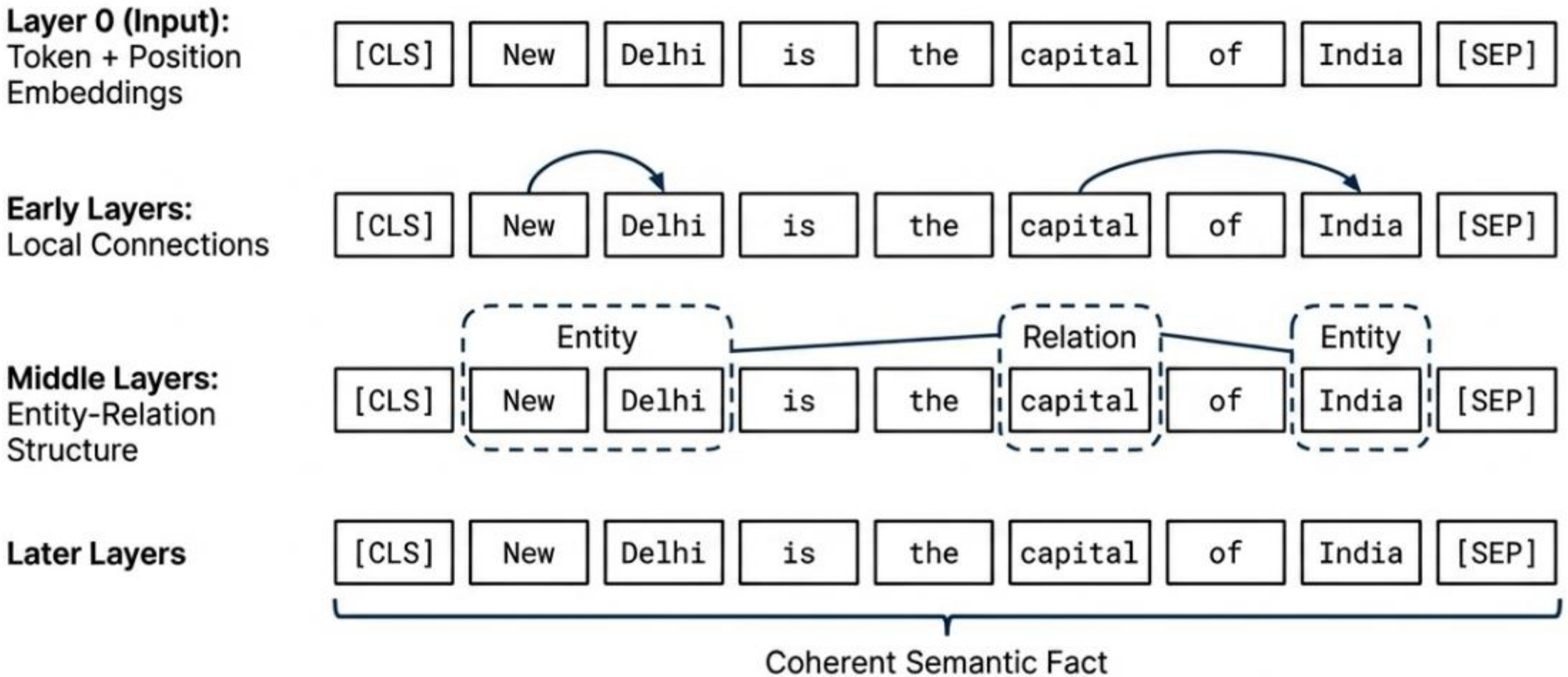
Positives (d+): Clicks, QA pairs, FAQ pairs, human annotations.

Negatives (d-): Random sampling, BM25, in-batch.

Note: Hard negatives are critical. They force the model to distinguish between superficially similar but actually irrelevant documents.

Capturing Semantic Meaning

Self-attention gradually transforms the model's understanding from isolated token identity and order into a fully contextualized sentence representation.

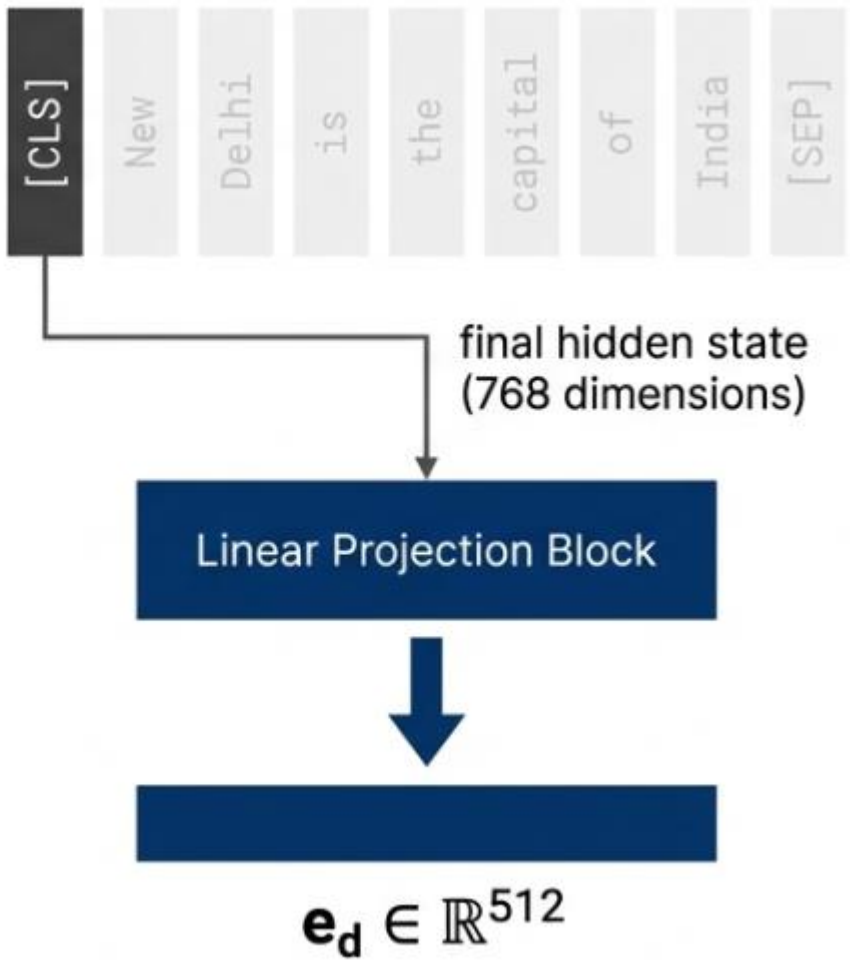


Lab: Visualizing Bi-Encoder Attention

https://tinyurl.com/rag-augmented

Compressing Context into a Single Representation

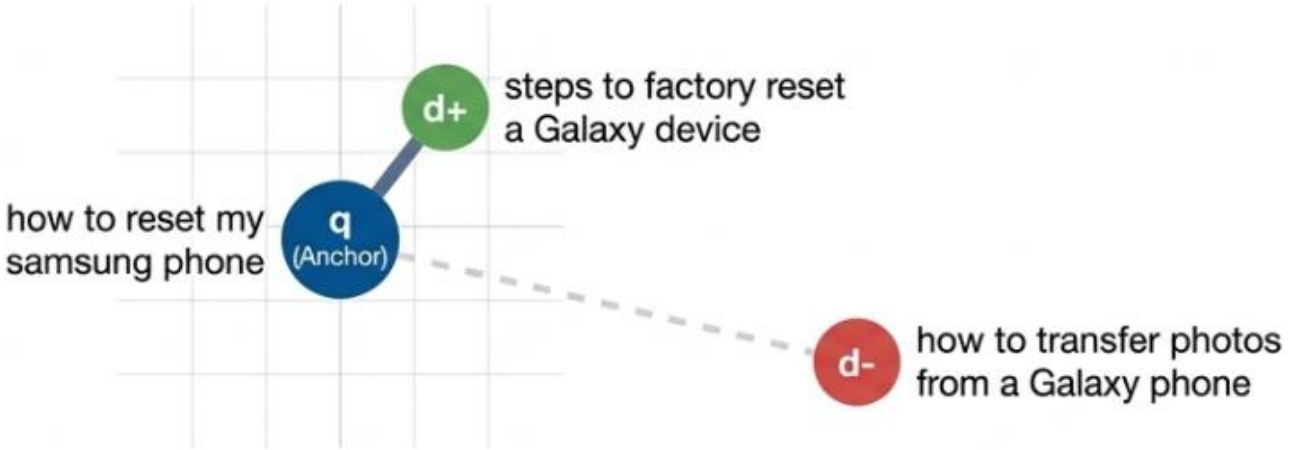
- Retrieval requires one vector for the entire sentence, not one per token.
- The final hidden state of the [CLS] token acts as a compact summary of the entire input.
- A linear projection layer maps the higher-dimensional hidden state down to a dense 512-dimensional embedding representing New Delhi is the capital of India.



Organizing the vector space by semantic intent

$$s(q, d+) > s(q, d-)$$

Rule: Relevant query-document pairs must be closer than irrelevant pairs.

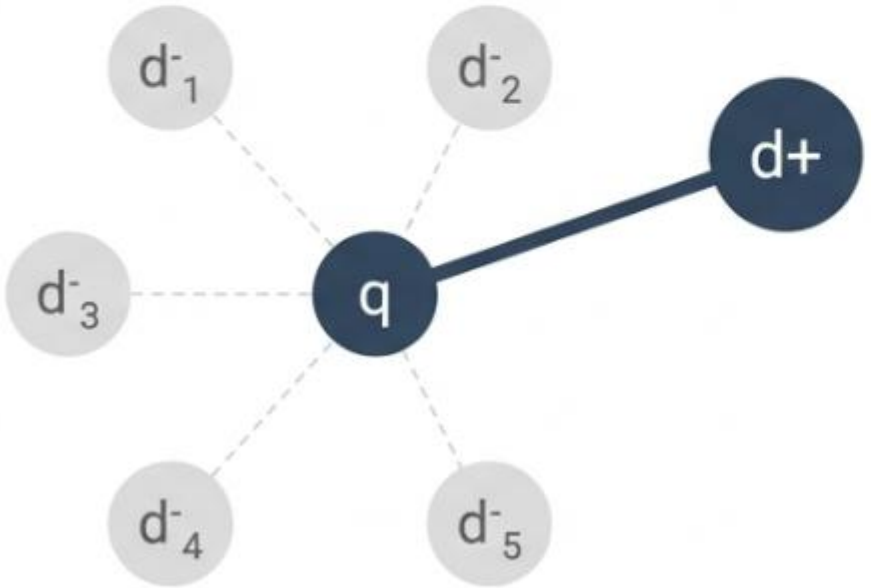


The model learns to map intent. Even though both documents mention 'Galaxy phone,' the vector space captures the true task-specific relevance over exact keyword matches.

InfoNCE Loss treats Retrieval as Classification

$$L = -\log\left[\frac{\exp(s(q,d^+)/t)}{\exp(s(q,d^+)/t) + \sum \exp(s(q,d_i^-)/t)}\right].$$

s(q,d) = similarity score, t = temperature



Core Intuition Details

Setup: One query (q), one positive (d+), several negatives (d^-_1, d^-_2, ...).

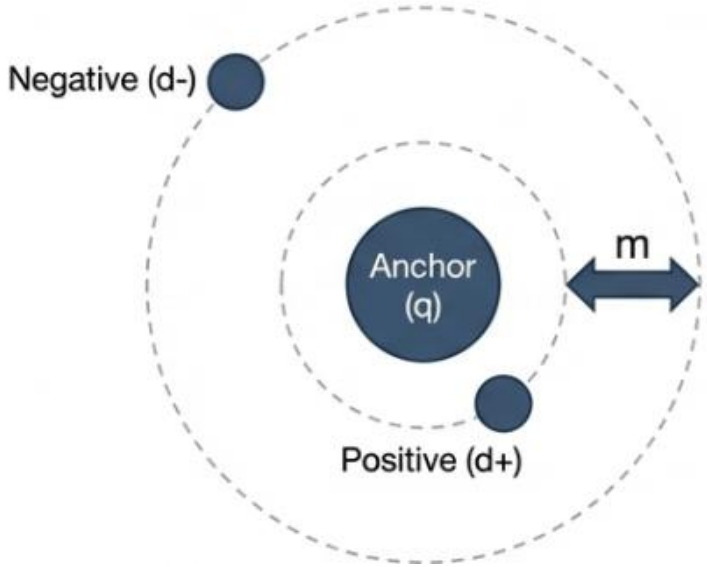
Mechanism: Softmax-based approach. Among all candidate docs, the model must assign the highest probability to the positive.

Advantage: Simple, effective, and works exceptionally well with in-batch negatives.

Triplet Loss Enforces a Spatial Margin

$$L = \max(0, D(q, d^+) - D(q, d^-) + m)$$

D = distance measure, m = margin. Alternatively: s(q,d+) >= s(q,d-) + m



Core Intuition Details

- **Anchor:** The query (q).
- **Positive:** Relevant document (d+) pulled close.
- **Negative:** Irrelevant document (d-) pushed away.
- **The Margin (m):** A clear visual gap enforced between the positive and negative distances.
- If the positive is already closer than the negative by at least the margin (m), the loss is zero and no penalty is applied.

Bi Encoder: Good and Bad

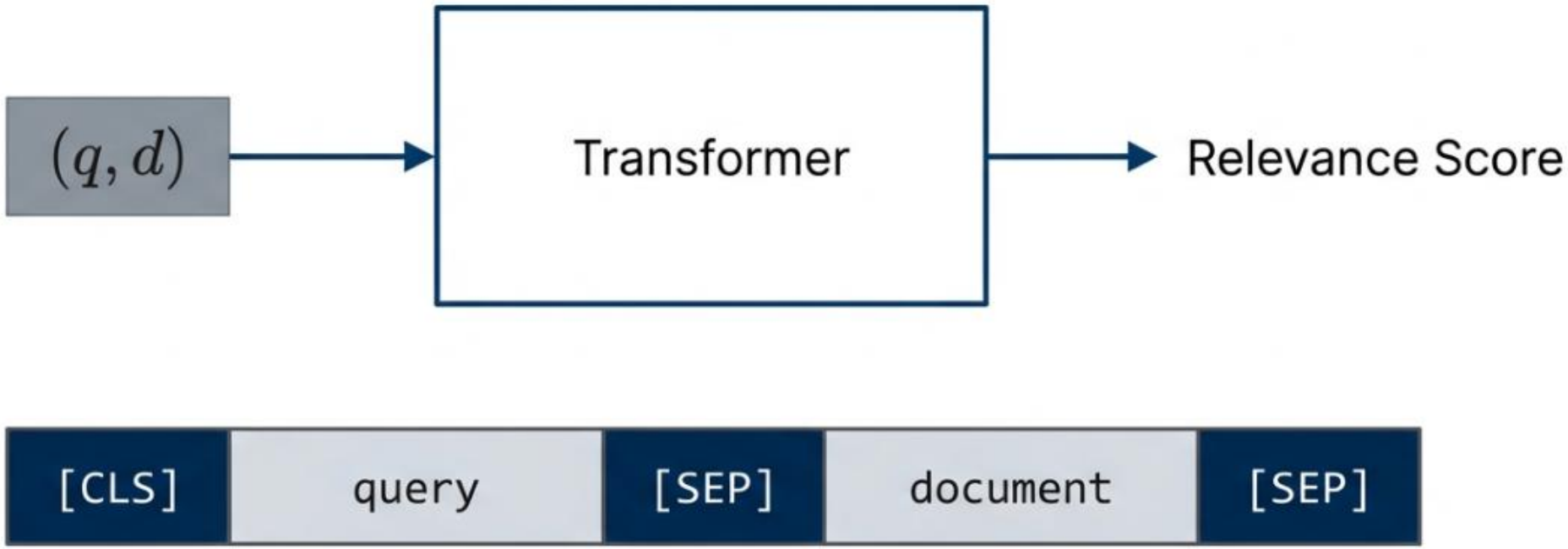
Strengths

- Scales to millions of documents.
- Lightning-fast runtime retrieval.
- Captures deep semantic similarity beyond exact keywords.

Limitations

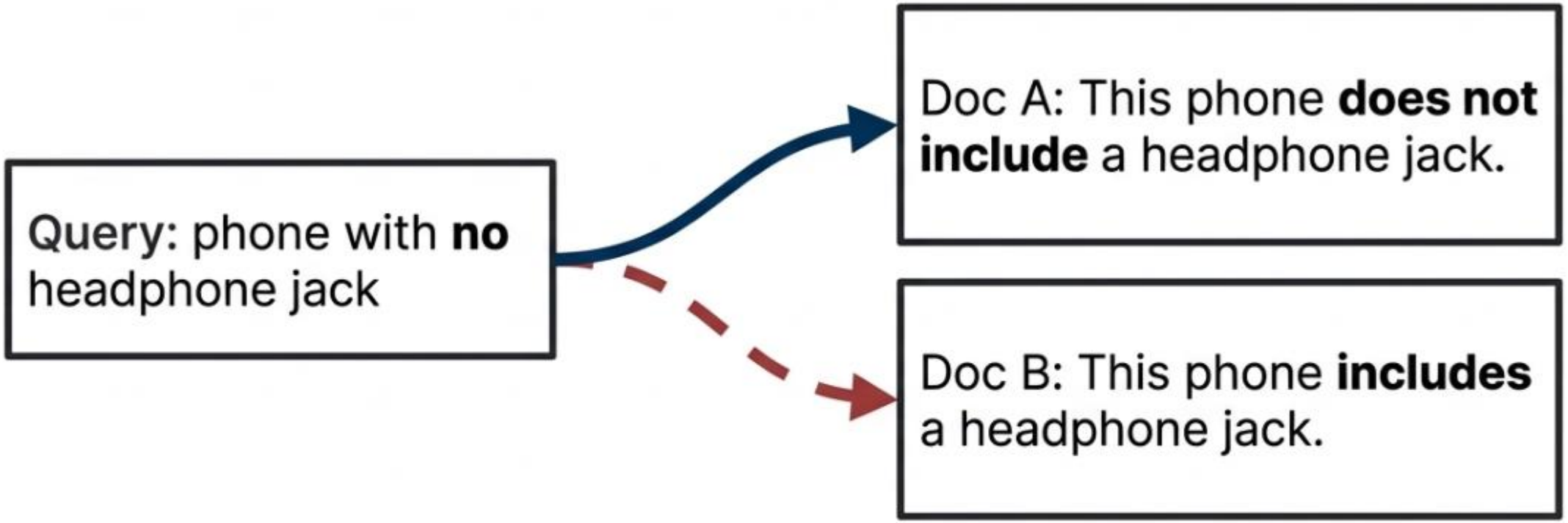
- No fine-grained token interaction at scoring time.
- Struggles with: negation, numerical constraints, exact entity disambiguation, and subtle wording differences.

Cross Encoder



Instead of creating isolated embeddings, the model jointly processes the query and document within the same Transformer. Every token in the query can interact directly with every token in the document.

Fine-Grained Comparison



Self-attention over the combined sequence allows query tokens to attend to document tokens and vice versa. The model asks: “Do these specific words and phrases match in the right way?”

Building Context Across Transformer Layers

Later Layers

- Capture overall relevance, query satisfaction, and condition checks.

Middle Layers

- Capture phrase-level alignment, semantic matches, and important constraints/modifiers.

Early Layers

- Capture local lexical patterns, nearby word relationships, and basic syntax.

[CLS] Token Aggregation

It is not just the query or just the document. It is an aggregated summary of the entire query-document interaction.

Vectors to Classifier

[CLS]_final → linear layer → z (raw score/logit) → $p = \sigma(z) = \frac{1}{1 + e^{-z}}$

p(q,d) = probability that document d is relevant to query q.	Score near 1 → Highly relevant. Score near 0 → Not relevant.

Cross-encoder reranking is framed as a binary relevance classification problem. Sigmoid forces the unbounded logit into a clean 0 to 1 probability space.

Binary Cross Entropy Loss

y = 1 (relevant pair)

$$L = -\left[y \log p(q,d) + (1-y)\log(1-p(q,d))\right]$$

Model's predicted probability

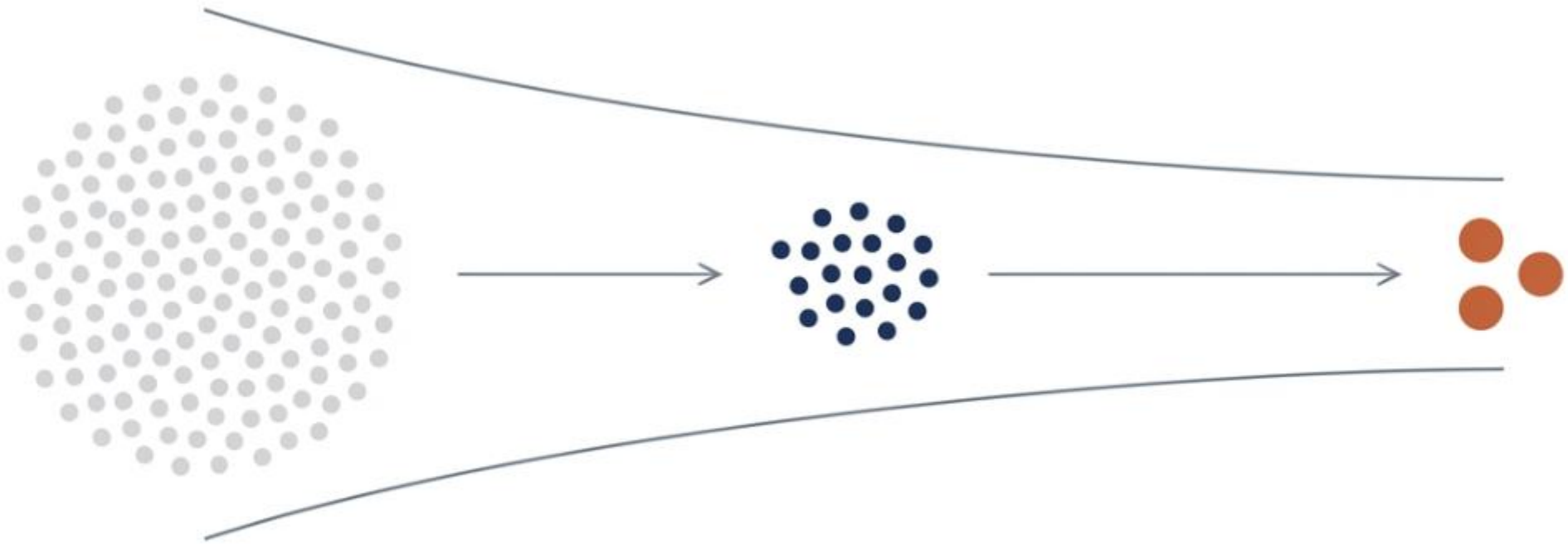
y = 0 (irrelevant pair)

Over millions of examples, the loss function penalizes the model if p(q,d) deviates from y. The model systematically learns which specific query-document token interactions signal relevance and which differences invalidate it.

Lab: Bi-Encoder and Cross-Encoder

https://tinyurl.com/rag-augmented

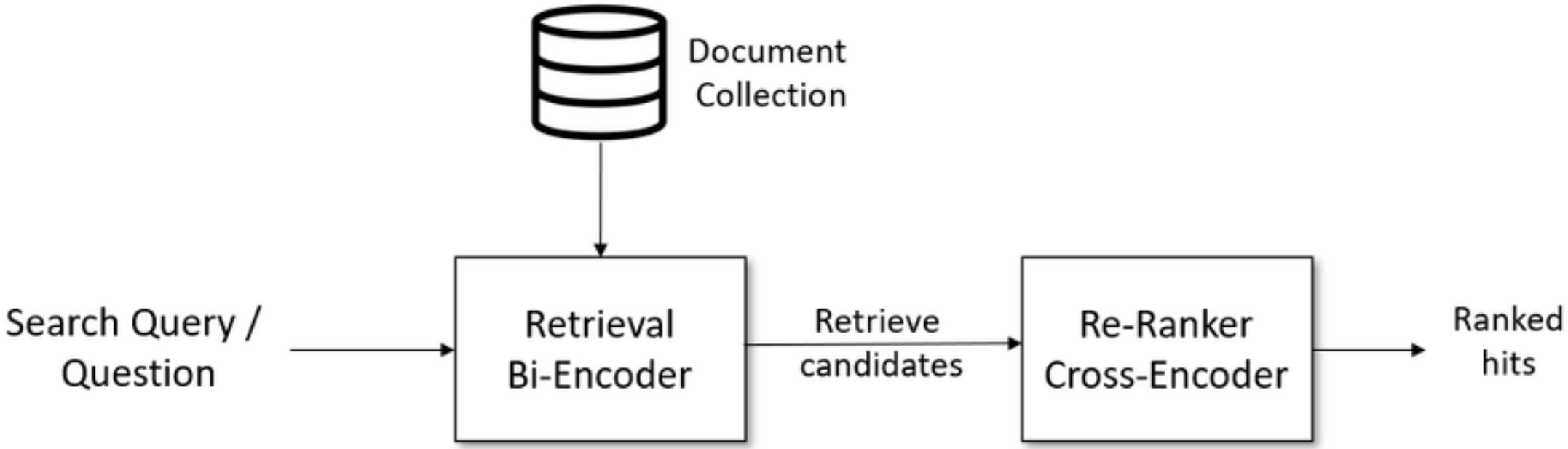
Retrieve-Rerank



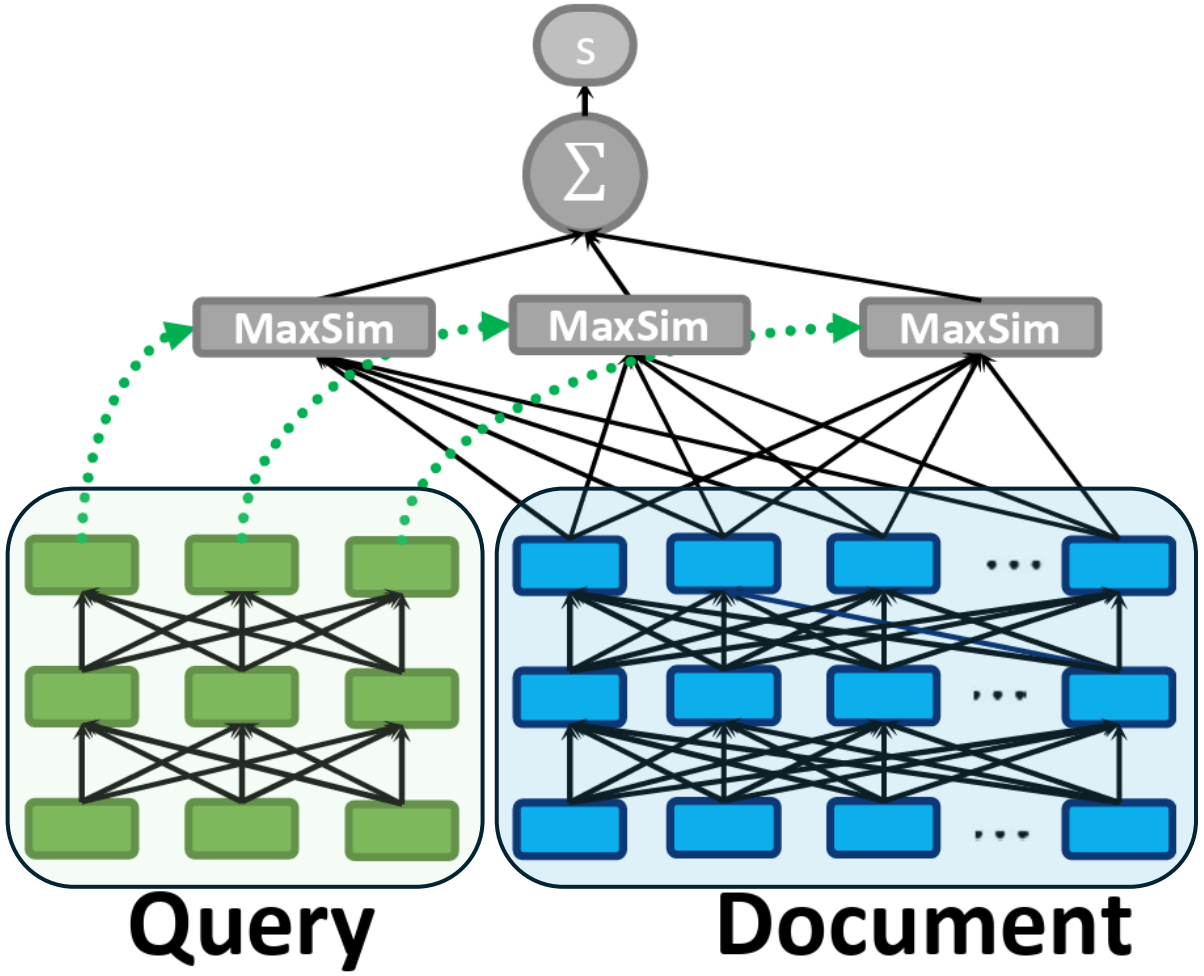
Retrieve candidate set from large corpus.

Rerank top candidates more accurately.

Pass top few to downstream RAG/User.


Document
Collection
Search Query /
Question
Retrieval
Bi-Encoder
Retrieve
candidates
Re-Ranker
Cross-Encoder
Ranked
hits

Late Interaction: ColBERT



Thank You

Appendix

A bi-encoder learns to place relevant query-document pairs close together in vector space.

Architecture

Encodes queries and documents separately, allowing massive offline precomputation.

Space

Learns a shared embedding space that prioritizes semantic intent.

Training

Fine-tuned via relevance supervision using InfoNCE or Triplet Loss.

Trade-off

Trades fine-grained token accuracy for unparalleled retrieval efficiency and scale.